

Dear examiner,

please find the criteria marks as well as teacher's comments below.

A: 4	B: 3	C: 2	D: 2	E: 5	Total: 16
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Criterion A: Presentation

An exploration that is easy to follow. The introduction has a clear aim, which is fully met. The structure is well-developed (introduction, body, conclusion); all sections link well to each other and the reader never feels lost. A number of equations and diagrams are used effectively to aid understanding. The exploration does not have any irrelevant or repetitive calculations.

Criterion B: Mathematical communication

The student uses relevant, appropriate mathematical communication for the most part. Equations, graphs, and diagrams are used effectively; key terms and variables are defined. A few minor slips are the incorrect use of the term "function" (page 4), using "inserting" instead of "substituting" (page 5), writing that "x" is a point (page 6), and a minor irregularity where she indicates that solving a definite integral is the same as evaluating a definite integral (page 6, "... the area under the graph can be expressed as: ...").

Criterion C: Personal engagement

The IA shows significant personal engagement. The student's voice can very much be heard. She took a creative approach presenting the concepts in a unique way and she incorporated a personal interest into the work. The IA contains lots of unfamiliar mathematics that is all self-taught and explained in a way that a fellow student can comprehend. The Math software GeoGebra is used effectively to draw graphs.

Criterion D: Reflection

Meaningful reflection is seen throughout the IA as well as at the end. The student successfully links back to the exploration's aim. In her conclusion the student summarizes what's been done. She discusses implications of the result, limitations to her modeling, and states a possible extension to her work. However, most of her reflection is not critical.

Criterion E: Use of mathematics

The mathematical arguments are clear and logical. The student brings together and utilizes a variety of relevant but unfamiliar mathematical concepts, such as two-variable functions, the equation for an ellipsoid, and solving double integrals to obtain volume. All mathematical working is error-free. At times the student is a bit sloppy with her mathematical language but this shortcoming was penalized in criterion B.

Finding the volume of the dome of the 'Museu Nacional' of Brasilia
using double integration

Student number: Hdv620

Pages: 20

Introduction, Rationale and Aim

In class we have explored in depth functions in a two-dimensional plane, with only one variable. We have investigated how to find the roots of the functions, understanding the elements of their graphs, and the meaning of their derivative and integrals. We were later introduced to the three-dimensional coordinate system in the vector's unit, which led me to question the behavior of functions in three dimensions, how functions with two variables worked. I then decided to do research into the concept of double integration of bi-variate functions. Because of the complexity and significance of integration I decided that it would be a topic worth exploring.

I was particularly interested in definite integrals due to their application in a one variable function, finding the area of a region between the graph and the x-axis. But what happens once we expand this concept into three dimensions? What can be calculated from it? It turns out that, the definite double integral of a function of two variables calculated over a region results in a volume (Lotha). Additionally, double integrals have a variety of applications in multiple areas. For example, to calculate density, mass and moment of inertia in Physics (Finan), or in Geography, to calculate average rainfall in an area (Strang). Furthermore, they are also used in probability to understand the relation between two random variables (Double Integrals in Probability). Although these go beyond the scope of this investigation, it is important to understand how mathematics, more specifically Calculus, serves a great utility in many areas of science.

I found the idea of finding the volume under a graph through integration in a three-dimensional coordinate system extremely interesting and useful and I decided to further investigate this using a building which I am familiar with and whose shape and structure have always interested me. This building is the 'Museu Nacional', a building that I used

to see everyday when I lived in Brasilia. I was always intrigued by its bowl shape and always wondered with a childish curiosity how much milk could you fit if you used the museum as a cereal bowl.



Figure 1: Museu Nacional de Brasília (Saldanha)

What is also interesting about this museum is that many sources such as the Wikipedia and even the Government of Brasilia website have mistakenly labeled the shape of the museum a semi-sphere or hemisphere which is an incorrect labeling since a hemisphere must also have an equal measurement for base radius, dome radius and height (Roberts). The dimensions of the museum, provided by the Government of Brasilia website, are dome with 25m of radius, a base of 32.6m of radius and a height of 26.25m (Secretaria De Estado De Cultura E Economia Criativa). This proves that the shape of the building has been mislabeled to simplify its description and the website is self-contradicting. This also shows that using the formula for the volume of a sphere is not a possible approach to finding the volume of the building and therefore double integration must be used.

The aim of this investigation is to understand the functioning and rules of double integration and its applications to apply it by discovering the volume of the aforementioned building. This will be done by explaining bi-variate functions and how to work with them. Followed by the steps of double integration, investigating cases with both rectangular and general regions. And finally, applying all findings to calculate the volume of the ‘Museu Nacional’.

Introduction to two variable equations

A function is considered multivariable if it has more than one input variable. A function can have an infinite number of variables, but in this investigation will only be concerned with bivariate equations, which have two variables. Such functions can have the graphs plotted in a three-dimensional space (figure 2)

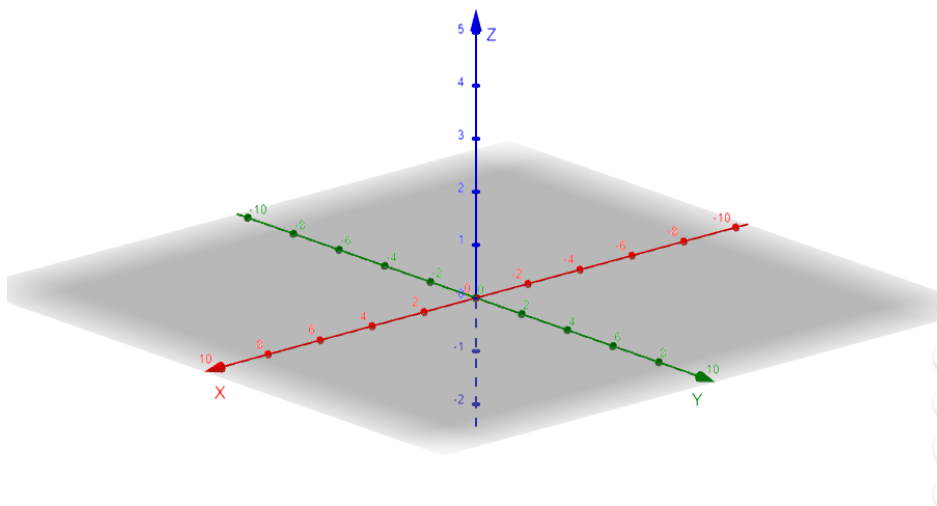


Figure 2: 3D coordinate grid (made with Geogebra)

A bi-variate function is denoted by $z = f(x, y)$, which for the simplification of this exploration: $x \in \mathbb{R}, y \in \mathbb{R}$ and $z \in \mathbb{R}$. In order to better understand the functioning

of bivariate equations we will investigate an example function. The equation chosen for this exploration is an ellipsoid which is a closed surface of which all plane cross sections are either ellipses or circles (Osserman). It has a general function:

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$$

In which a, b and c are constants and positive real numbers. Now rearranging the function for z :

$$\frac{z^2}{c^2} = 1 - \frac{x^2}{a^2} - \frac{y^2}{b^2}$$

$$z^2 = c^2 \left(1 - \frac{x^2}{a^2} - \frac{y^2}{b^2}\right)$$

$$z = \pm \sqrt{c^2 \left(1 - \frac{x^2}{a^2} - \frac{y^2}{b^2}\right)}$$

The reason for choosing this function is because the shape of the building whose volume I intend to find is a section of an ellipsoid. Therefore, it is important to investigate its graph. Figure 4 represents the graph of an ellipsoid in which all constants, a, b and c are equal to 1, forming a sphere.

$$z = \sqrt{1 - x^2 - y^2} \text{ and } z = -\sqrt{1 - x^2 - y^2}$$

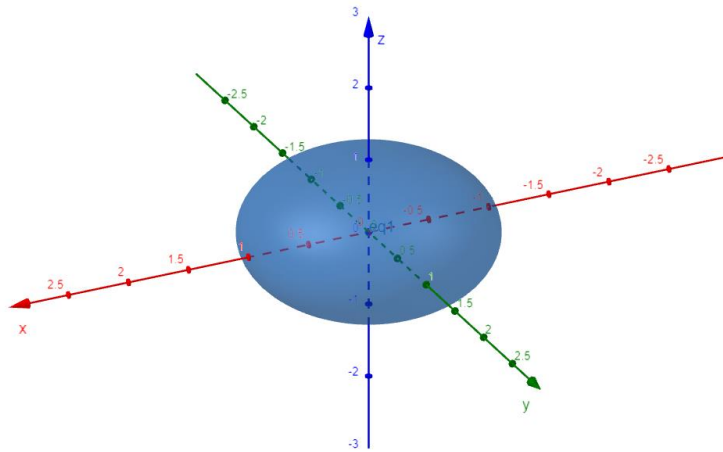


Figure 3: Graph of equation $z = \pm\sqrt{1 - x^2 - y^2}$ (made with GeoGebra)

The equation has domain $x, y \in (-1, 1)$ and $x^2 + y^2 < 1$, and range $z \in (-1, 1)$. By limiting the range, $z \in (0, 1)$ we have a function which values of z can be found by inserting two inputs, for example:

$$(x, y) = \left(\frac{1}{2}, -\frac{1}{2}\right)$$

$$z = \sqrt{1 - \left(\frac{1}{2}\right)^2 - \left(-\frac{1}{2}\right)^2}$$

$$z = \sqrt{1 - \frac{1}{4} - \frac{1}{4}}$$

$$z = \sqrt{0.5}$$

$$z \approx 0.707$$

It is significant to mention that not all inputs of x and y will generate a real output, if $x = \pm 1$ and $y = \pm 1$ then the result for z will be complex.

$$z = \sqrt{1 - (\pm 1)^2 - (\pm 1)^2}$$

$$z = \sqrt{1 - 1 - 1}$$

$$z = \sqrt{-1}$$

Integrate two variate functions.

In order to understand how integration of two variables works, it is necessary to revise some concepts of integration in the x,y plane. We know that to find the area between the graph of a function and the x-axis, one can use the definite integral of the function. Which leads to the sum of the area of infinitely small rectangles (see Fig 4) under the graph in a delimited region. Below is the equation of the definite integral:

$$\int_a^b f(x)dx = [F(b)] - [F(a)]$$

a and b are constant values of x , and the lower and upper boundaries of the region under the graph. $F(x)$ is the integral of $f(x)$, and dx is an infinitely small change in x , or the base of each rectangle. It can also be said that y is the height of such rectangles. The interval is divided into n subintervals of width dx , and from each subinterval a point x is chosen. Then the sum of the infinitely small rectangles, and the area under the graph can be expressed as:

$$\lim_{n \rightarrow \infty} \sum_{n=1}^{\infty} f'(x)dx = \int_a^b f(x) dx$$

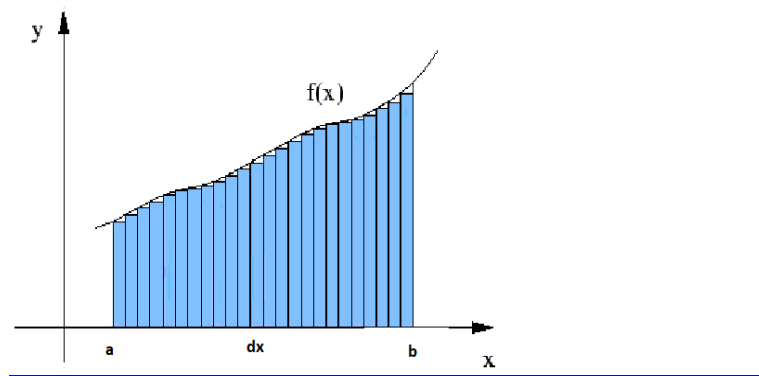


Figure 4: Definite integral in the xy coordinate system (pleacher)

We can apply this same principle in a three-dimensional coordinate system when trying to discover the volume between the graph of a function and the xy plane. The first thing to consider are the boundaries necessary to find the definite integral (as seen in Fig 5) four boundaries are necessary.

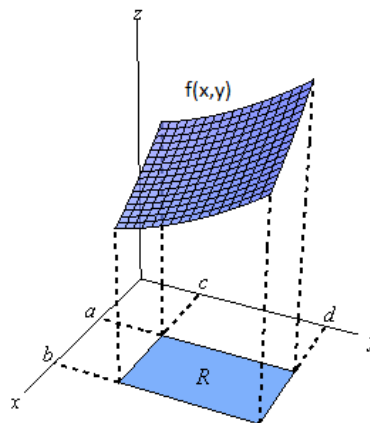


Figure 5: Double integration over a rectangular area (Dawkins)

The boundaries on the y -axis will be denoted as c and d , where $c \in \mathbb{R}$ and $d \in \mathbb{R}$.

The first step in understanding double integration is by considering a constant coordinate in y , for example point d . In this case y will be treated as a constant and we

create a slice (in red, Fig 6) under the function $f(x, y)$, and we can use integration with only one variable, x , to find the area of the slice.

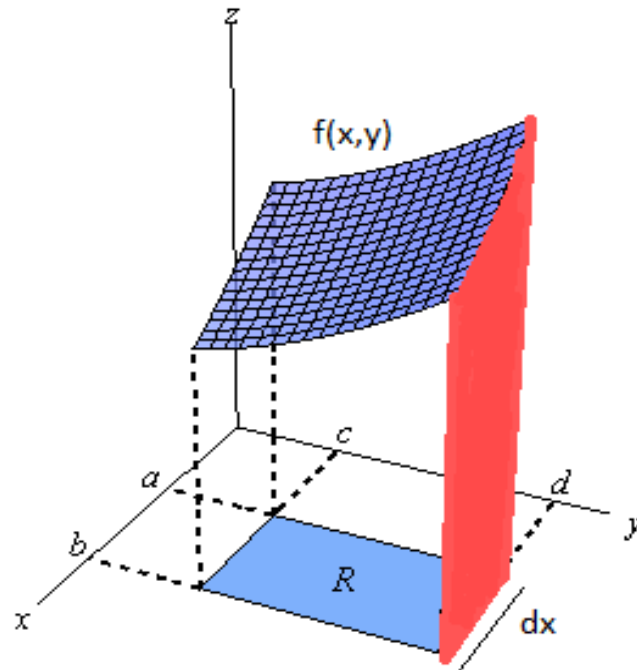


Figure 6: Double integration over a rectangular area with self-made annotations (Dawkins)

By dividing this area into infinitely small rectangles with height z and base dx . Therefore, its area can be calculated with an integral in respect to x . Leading to the equation for the area of slice for any given y :

$$A(\text{slice}) = \int_a^b f(x, y) dx$$

But in order to find the volume of the region it is necessary to multiply this area by the sum of an infinitely small change in y , within the chosen boundaries. Therefore,

$$V = \int_c^d \int_a^b f(x, y) dx dy$$

Definite integration over variable regions

The previous sections investigated finding the volume of regions between the graph of a bivariate function and the xy plane that have a rectangular base. Meaning that the boundaries a, b, c and d , are all constants. Now the exploration will investigate finding the volumes in situations in which the region created on the xy plane is not rectangular. Since the boundaries are now variables and no longer constants they will be denoted as followed

$$V = \int_{y_1}^{y_2} \int_{x_1}^{x_2} f(x, y) \, dx \, dy$$

The region can be any shape created by a function $y = f(x)$ bounded by a specific domain. For example, if the area is a triangle created the function $y = x$, when $0 \leq x \leq 2$, then it can be graphed in the xy coordinate (Fig 7)

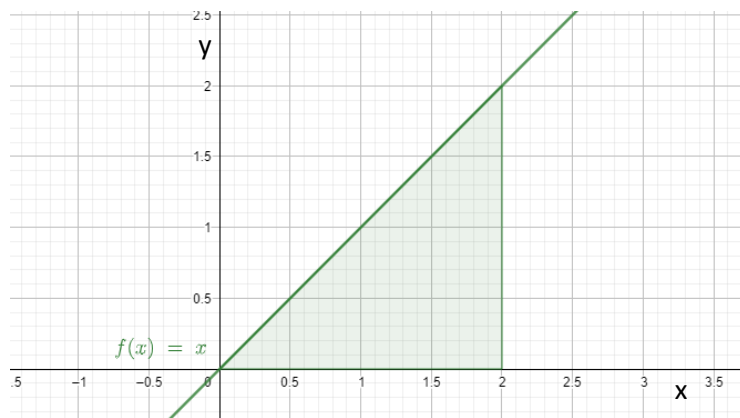


Figure 7: Graph of $f(x)=x$ (made with GeoGebra)

Going back to the concept of finding the area of slices of the region, in the case of a non-rectangular region not only the height of the slices will change (based on the graph of $z = f(x, y)$) but the length of their base also changes (based on the function $y = f(x)$).

Considering the example given: when $x = 1$, y will range from 0 to 1 and if $x = 2$, then y ranges from 0 to 2.

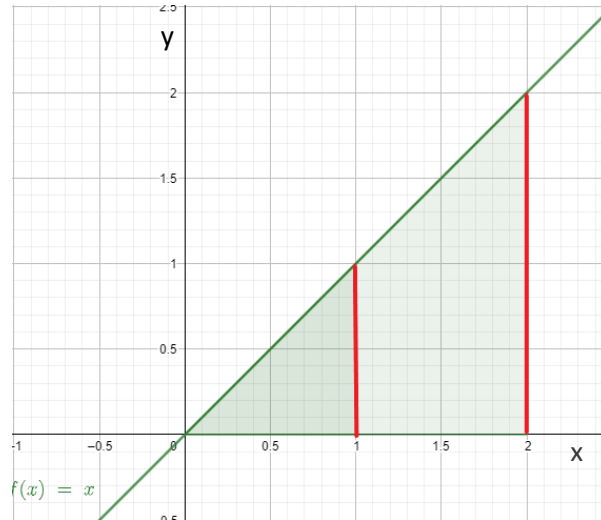


Figure 8: Graph of $f(x)=x$ with annotations (made with Geogebra)

In order to express this change, the boundaries of the inner integral have to be the function with respect to the outer variable (“Double Integrals over Non-Rectangular Regions (Article).”)

$$V = \int_{y_1}^{y_2} \int_{x_1=f(y)}^{x_2=f(y)} f(x, y) \, dx \, dy$$

This will lead to the evaluation of the inner integral being a function of y , which can later be integrated in respect to y .

Looking back at the example in which the function of the area in the xy plane is the equation $y = x$, then we can substitute the boundaries of the inner integral for a function $x = f(y)$ to achieve:

$$V = \int_{y_1}^{y_2} \int_{x=y}^{x=y} f(x, y) dx dy$$

Solving the opening problem:

Now I will use the newfound knowledge to solve the opening problem and finding the volume of the ‘Museu Nacional’ building in Brasilia. For purpose of simplification the volume calculated will only be of the museum’s dome, without taking into consideration the intersecting walkways. Since the government source that provides dimensions for the museum is self-contradicting, I decided to research into more reliable sources. With that, I found schematic images of the museum’s dome with different dimensions from the other source (Fig 9). The source is the department of Architecture in the University of Brasilia (Inojosa), which is more reliable due to the expertise in the area.

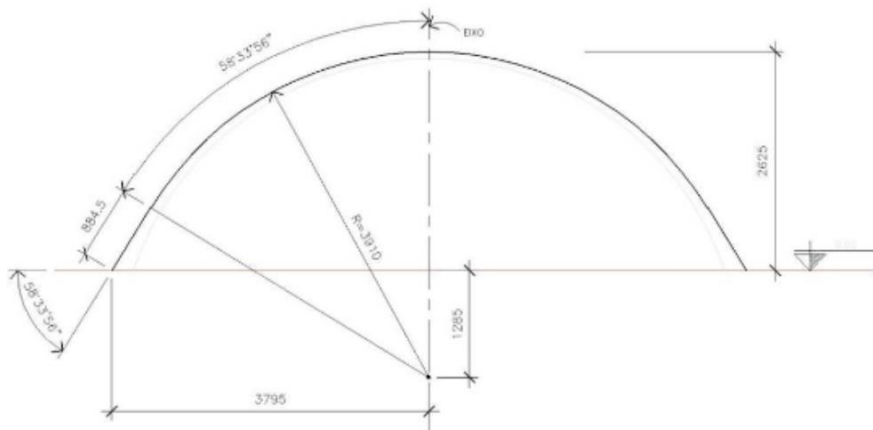


Figure 9: Dimensions of dome (Inojosa)

This means that the museum’s dimensions truly are dome with 39.10m of radius, a base of 37.95m of radius and a height of 26.25m

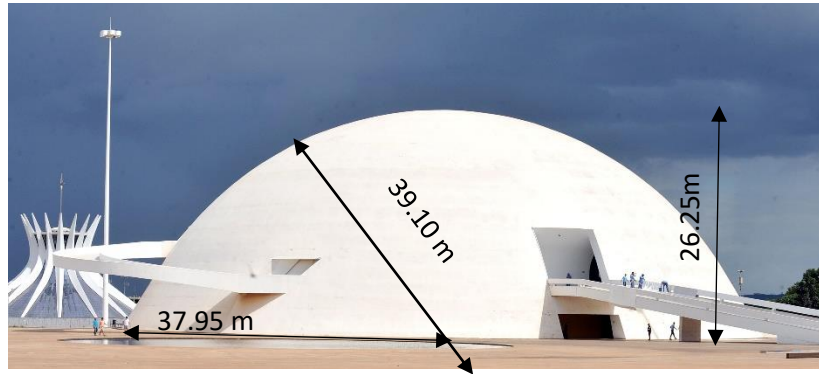


Figure 10: Museu Nacional with self-made annotations (Apsil)

From this information we can assume that the base of the building is a circle with radius 37.95 m (see Fig 10)

The first step is finding a function $z = f(x, y)$, that models the dome of the building. As previously mentioned, the dome of the museum is the section of an ellipsoid, because its planes cross-sections are either ellipses or circles. Therefore, the function of the dome will be derived from the general function of an ellipsoid:

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$$

Or,

$$z = \pm \sqrt{c^2 \left(1 - \frac{x^2}{a^2} - \frac{y^2}{b^2} \right)}$$

The next step is finding the values for a , b and c . In order to find it one must understand the properties of an ellipsoid, the ellipsoid has three perpendicular axes of symmetry that intersect at the center of symmetry of the ellipsoid, these are called the principal axes (“Ellipsoid”). The axes can be the x , y and z axes (Fig 11).

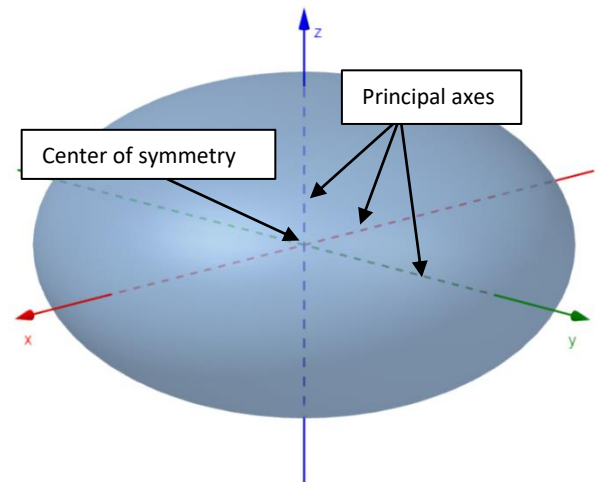


Figure 11: Graph of an ellipsoid with annotations, made with GeoGebra

Furthermore, the points $(a, 0, 0)$, $(0, b, 0)$ and $(0, 0, c)$ lie on the surface and the line segments from the origin to these points are the principal semi-axes, or half of the principal axes (“Ellipsoid”). Therefore, a is the distance from the origin to the x -intersect of the graph, b is distance from origin to the y -intersect and, c is distance from origin to z -intersect (Fig 12).

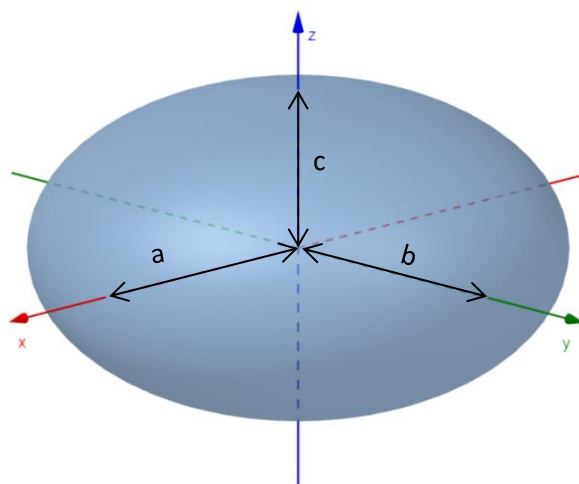


Figure 12: Graph of an ellipsoid with annotations, made with GeoGebra

Now applying the findings to the museum's dome dimensions, we have placed the base of the museum in the xy-plane, and it is known that this base is a perfect circle with radius of 37.95 m, therefore it can be inferred that $a = b = 37.95$. As for c , it is equivalent to the radius of the dome, so $c = 39.10$. This leads to the function.

$$z = \pm \sqrt{39.10^2 \left(1 - \frac{x^2}{37.95^2} - \frac{y^2}{37.95^2}\right)}$$

When plotting the equation and cutting it in the xy plane, a dome that has a radius of 39.10m and circular base of radius 37.95m is created (Fig 13). These dimensions agree to the dimensions of the museum. But the height of such dome is 39.10 m, while the museum has height 26.25m

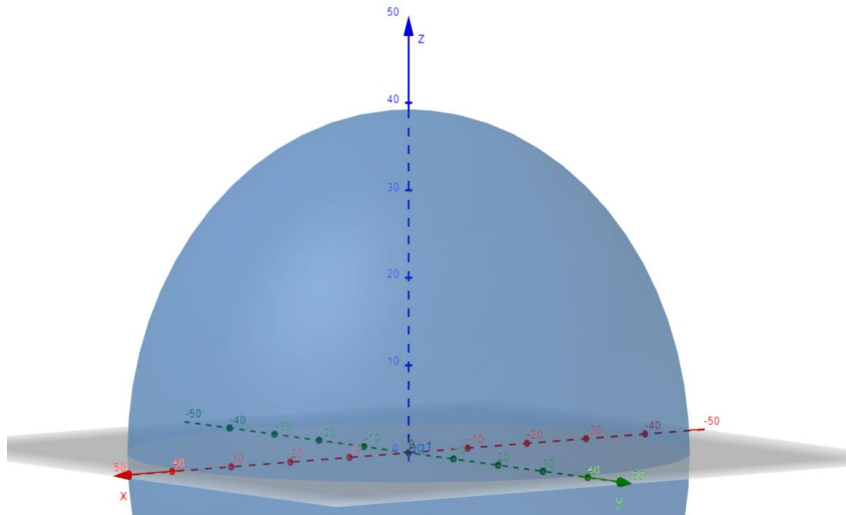


Figure 13: Graph of dome equation (made with GeoGebra)

This means that the museum is less than half of an ellipsoid. This also means that the graph must be translated n meters downwards in the z -axis to find get the desired height. Therefore, n meters must be subtracted from the equation.

$$z = \pm \sqrt{39.10^2 \left(1 - \frac{x^2}{37.95^2} - \frac{y^2}{37.95^2}\right)} - n$$

The amount of n meters to be translated is equal to the height of the graph minus the desired height: $n = 39.10 - 26.25$.

$$n = 12.85$$

Hence, the equation of the dome is:

$$z = \pm \sqrt{39.10^2 \left(1 - \frac{x^2}{37.95^2} - \frac{y^2}{37.95^2}\right)} - 12.85$$

Now graphing the new equation:

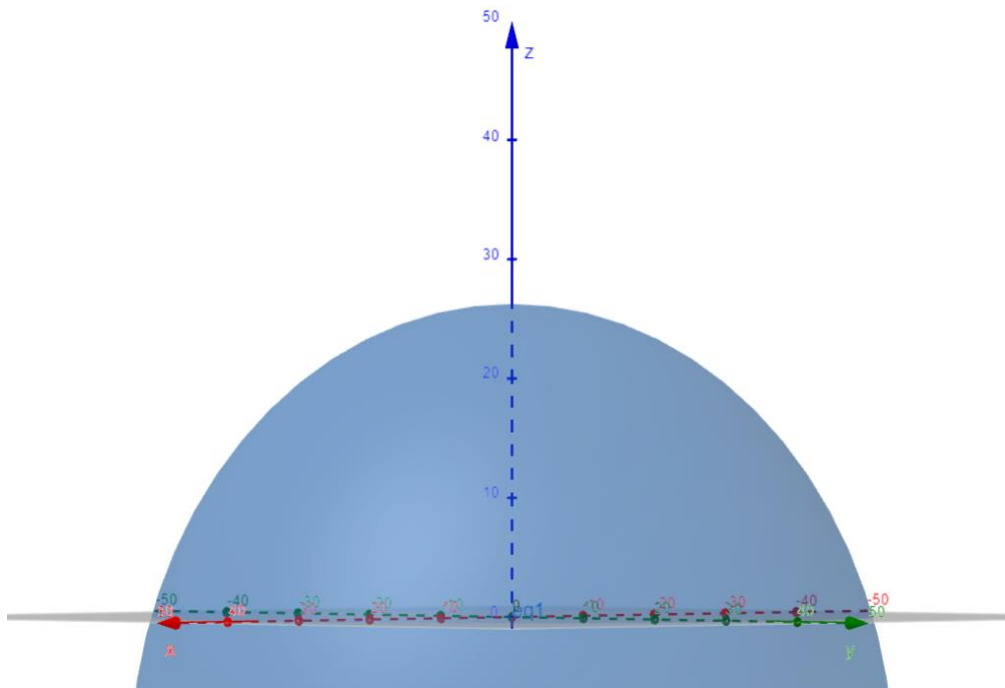


Figure 14: Final graph of dome equation (made with GeoGebra)

As the base of the dome is the xy plane, the dome is represented by only positive values of z, so the final equation is:

$$z = \sqrt{39.10^2 \left(1 - \frac{x^2}{37.95^2} - \frac{y^2}{37.95^2}\right)} - 12.85, \quad z \in \mathbb{R}^+$$

When the graph is translated upwards the distances between the origin and the x-intercepts, and the y-intercepts decrease from 37.95m to 35.85m. This means that the dome of the museum is not a perfect section of an ellipsoid, but rather has its sides are extended about 2.10 m. But in order to facilitate this investigation we will not consider this extension. Therefore, the new radius of the base to be considered is 35.85 meters.



Figure 15: Museu Nacional seen from above with annotations (made with Google Maps)

The base of the dome can be graphed in the xy coordinate system using the general function of a circle, where r = radius

$$x^2 + y^2 = r^2$$

Substituting r for 35.85 m

$$x^2 + y^2 = 35.85^2$$

Below is the graph for the function:

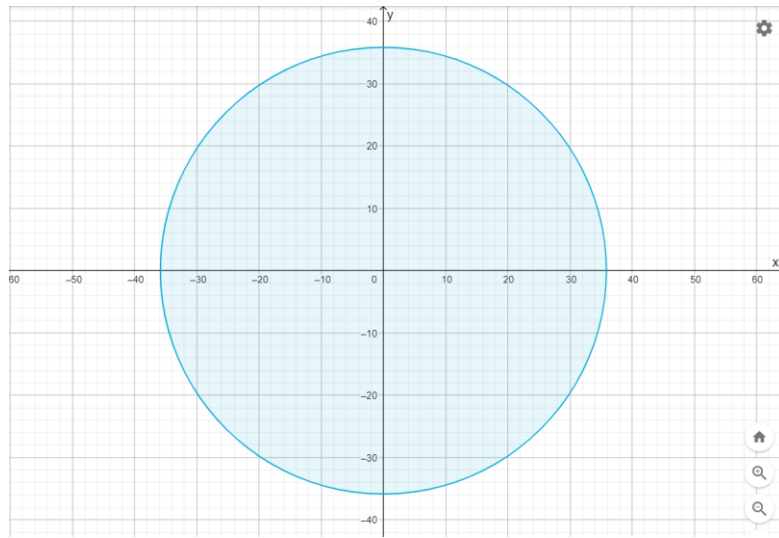


Figure 16: graph of circle (made with GeoGebra)

Now this information can be used to find the lower and upper boundaries of integration. The volume of the dome of the museum will be a definite double integral of the function defined within this circular region on the xy plane. The formula for definite double integration over a variable region for the function of the dome is:

$$V(\text{dome}) = \int_{y_1}^{y_2} \int_{x_1=f(y)}^{x_2=f(y)} \sqrt{39.10^2 \left(1 - \frac{x^2}{37.95^2} - \frac{y^2}{37.95^2} \right) - 12.85} \, dx \, dy$$

The next step is finding the boundaries of the inner integral which must be in the format of $x = f(y)$. So, by simply adjusting the equation of the circle:

$$x^2 + y^2 = 35.85^2$$

$$x^2 = 35.85^2 - y^2$$

$$x = \pm \sqrt{(35.85^2 - y^2)}$$

Now, when considering any slice in the region, which will be denoted by $\overline{ST}$ (figure 17) it can be affirmed that the coordinates of S and T will always be:

$$S = (-\sqrt{35.85^2 - y^2}, y) \text{ and } T = (\sqrt{35.85^2 - y^2}, y)$$

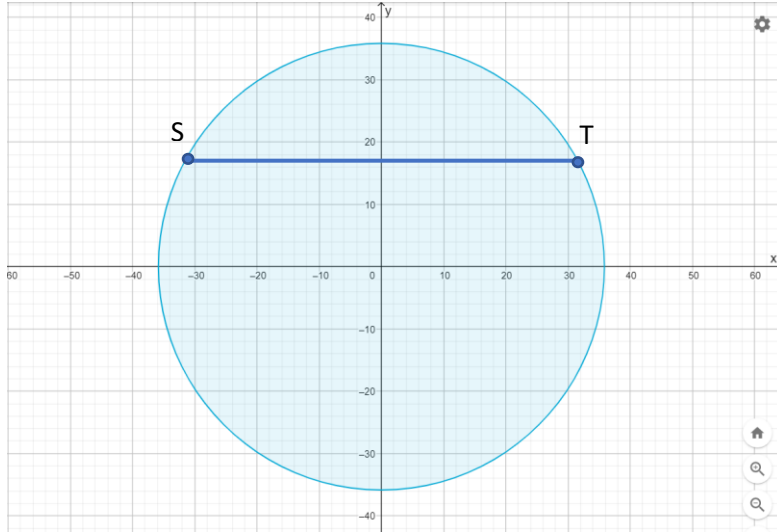


Figure 17: Graph of circle with annotations (made with GeoGebra)

Finally, we can evaluate the inner integral using the x coordinates for points S and T as the lower and upper boundaries for the inner integral. As for the boundaries of the outer integral, it is the y-coordinates of the y-intercepts:

$$V(dome) = \int_{-35.85}^{35.85} \int_{-\sqrt{35.85^2 - y^2}}^{\sqrt{35.85^2 - y^2}} f(x, y) dx dy$$

Now we can integrate this equation with the boundaries found to find the volume of the museum's dome:

$$V(dome) = \int_{y_1}^{y_2} \int_{-\sqrt{35.85^2 - y^2}}^{\sqrt{35.85^2 - y^2}} \left(\sqrt{39.10^2 \left(1 - \frac{x^2}{37.95^2} - \frac{y^2}{37.95^2} \right)} - 12.85 \right) dx dy$$

First, we integrate with respect to x , treating y as a constant. Due to its complexity, the integral will be calculated using the website [integral-calculator.com](https://www.integral-calculator.com) (Scherfgen)

$$\int_{-\sqrt{35.85^2-y^2}}^{\sqrt{35.85^2-y^2}} \left(\sqrt{39.10^2 \left(1 - \frac{x^2}{37.95^2} - \frac{y^2}{37.95^2} \right)} - 12.85 \right) dx$$

$$= \frac{(979337 - 6800y^2) \times \sin^{-1} \left(\frac{\sqrt{400y^2 - 514089}}{\sqrt{400y^2 - 576081}} \right) + (102\sqrt{1722} - 8481) \times i\sqrt{400y^2 - 514089}}{6600}$$

The next step is to calculate the outer integral. Again, the website [integral-calculator.com](https://www.integral-calculator.com) (Scherfgen) will be used.

$$\int_{-35.85}^{35.85} \left(\frac{(979337 - 6800y^2) \times \sin^{-1} \left(\frac{\sqrt{400y^2 - 514089}}{\sqrt{400y^2 - 576081}} \right) + (102\sqrt{1722} - 8481) \times i\sqrt{400y^2 - 514089}}{6600} \right) dy$$

$$\approx 1.1387163427474 \times 10^5$$

Hence,

$$V(\text{dome}) \approx 1.1387163427474 \times 10^5 \text{ m}^3$$

Conclusion and Reflection

This investigation has explained how to integrate two-variate functions, in both triangular and variable regions, and applied these findings to find the volume of the dome of the ‘Museu Nacional’ in Brasilia. This finding could have implications in the area of

architecture or for the management of the museum, for example, when installing a ventilation system or delimiting the maximum capacity of people in the museum, which is especially important to avoid the spread of diseases that are spread through aerosol particles.

Even though the investigation has found a reliable result for the volume, there are still some limitations such as the previously mentioned change of 2.10 meters of the radius of the base and the fact that this investigation does not take into account the thickness of the concrete used to build the museum. If this investigation were to be extended to have a more reliable result, considering the extension of the base, it would be necessary to find a function that models the extension of the base and integrate it in a ring-shaped base.

Works Cited

- “3D Calculator.” *GeoGebra*, www.geogebra.org/3d.
- Apasil, Milton. “Museu Nacional - Brasília, Brasil - Oscar Niemeyer.” *Pinterest*, www.pinterest.com/pin/390405861420713525/.
- Dawkins, Paul. *Calculus III - Double Integrals*, 2003, tutorial.math.lamar.edu/classes/calciiii/DoubleIntegrals.aspx.
- Double Integrals in Probability*, web.ma.utexas.edu/users/m408s/m408d/CurrentWeb/LM15-5b-3.php.
- “Double Integrals over Non-Rectangular Regions (Article).” *Khan Academy*, Khan Academy, www.khanacademy.org/math/multivariable-calculus/integrating-multivariable-functions/double-integrals-a/a/double-integrals-over-non-rectangular-regions.
- “Ellipsoid.” *Wikipedia*, Wikimedia Foundation, 21 Mar. 2021, en.wikipedia.org/wiki/Ellipsoid#:~:text=An%20ellipsoid%20is%20a%20surface,degree%20two%20in%20three%20variables.
- Finan, Marcel B. “Applications of Double Integrals.” Arkansas Tech University.
- Index of /Mp/Mgifs/gifs8*, www.pleacher.com/mp/mgifs/gifs8/.
- Inojosa, Leonardo, et al. *Aspectos Do Projeto Estrutural Do Museu Nacional De Brasília*, Universidade Federal de Brasília, docplayer.com.br/1916902-Aspectos-do-projeto-estrutural-do-museu-nacional-de-brasilia.html.
- Lotha, Gloria. “Multiple Integral.” *Encyclopædia Britannica*, Encyclopædia Britannica, Inc., 2010, www.britannica.com/science/multiple-integral.
- Osserman, Robert. “Ellipsoid.” *Encyclopædia Britannica*, Encyclopædia Britannica, Inc., 25 Aug. 2006, www.britannica.com/science/ellipsoid.
- Roberts, Donna. *Spheres and Hemispheres - MathBitsNotebook(Geo - CCSS Math)*, 2021, mathbitsnotebook.com/Geometry/3DShapes/3DSpheres.html.
- Saldanha, Francisco Willian. “Museu Nacional De Brasília.” *Wikimedia Commons*, 11 June 2017, commons.wikimedia.org/wiki/File:Museu_Nacional_de_Bras%C3%ADlia_3.jpg.
- Scherfgen, David. “Integral Calculator.” *Integral Calculator • With Steps!*, 2021, www.integral-calculator.com/.

Secretaria De Estado De Cultura E Economia Criativa. “Museu Nacional.”
SECRETARIA DE ESTADO DE CULTURA E ECONOMIA CRIATIVA,
www.cultura.df.gov.br/museu-nacional/.

Strang, Gilbert. “15.4: Applications of Double Integrals.” *Mathematics LibreTexts*,
Libretexts, 10 Nov. 2020,
[math.libretexts.org/Bookshelves/Calculus/Map%3A_Calculus__Early_Transcendentals_\(Stewart\)/15%3A_Multiple_Integrals/15.04%3A_Applications_of_Double_Integrals](https://math.libretexts.org/Bookshelves/Calculus/Map%3A_Calculus__Early_Transcendentals_(Stewart)/15%3A_Multiple_Integrals/15.04%3A_Applications_of_Double_Integrals).